

EMG Analysis of Bicep Curls at Different Angles

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Summary

This study investigates how changes in shoulder and trunk angle influence biceps brachii activation during dumbbell biceps curls. Surface electromyography was used to measure activity in the long head and short head while one participant performed curls at three orientations, which were zero degrees, forty-five degrees, and fifty degrees. The EMG signals were filtered to obtain the linear envelope, and mean, peak, and root mean square values were calculated for each condition. The fifty degree condition produced the highest peak and root mean square values, while the zero degree and forty five degree conditions showed lower overall activation. The findings suggest that a more reclined trunk and shoulder position increases biceps activation during the curl movement.

Introduction

The biceps brachii is an important muscle for upper limb movement and plays key roles in elbow flexion, forearm supination, and support of the glenohumeral joint [4]. It contains two heads, the long head and the short head, which differ in anatomical position and mechanical contribution during arm motion [4]. Dumbbell biceps curls are commonly used in resistance training and serve as a practical model for studying neuromuscular behavior during elbow flexion [2]. Although many studies have examined factors such as elbow angle and grip, fewer have explored how shoulder position influences recruitment of the two heads during a curl [3]. Changes in shoulder angle modify muscle length and leverage, which can shift the point where each head reaches its highest activation [5]. Prior research comparing different curl variations shows that activation of the long head depends on shoulder position, with extended positions maintaining higher activation and flexed positions emphasizing early phase activation [5]. Additional work indicates that altering glenohumeral angle across sets increases overall biceps activation [1], and electromyography studies confirm that small setup changes can affect activation timing and magnitude [2].

A clear understanding of how activation changes with shoulder position can guide exercise selection and support more effective strengthening programs [3]. This study examines how shoulder angles of zero degrees, 45 degrees, and 50 degrees influence activation of the long head and the short head of the biceps brachii during a dumbbell curl. Surface electromyography is used to measure muscle activity, and the analysis focuses on mean activation, peak activation, and root mean square values across the curl. This provides a descriptive comparison of activation patterns and identifies the shoulder position that produces the highest level of normalized muscle activity.

Methods

Through this experiment, we used surface electromyography (EMG) to investigate the effect of arm and trunk angle on the biceps muscle during bicep curls. Prior to electrode placement, the participant's skin was first cleaned using alcohol wipes to reduce signal noise during the measurement and electrical impedance. After proper skin preparation, surface electrodes were placed over the belly of the biceps and a reference electrode (ground) was placed in a neutral bony landmark at the elbow. The EMG was connected to a BioRadio and a constant external load of approximately 3 kg was used throughout the experiment. The participant was seated on a bench for all conditions to minimize body movement and isolate elbow flexion. We used three different angle orientations: an upright seated at positive 45° relative to the vertical, a declined position with the participant leaning back at 50°, and a neutral position with the forearm aligned with the positive z-axis of the 3D coordinate system. For each condition, the participant performed controlled bicep curl repetitions consisting of flexion and extension at a steady and constant pace to minimize the use of momentum. We then filtered the collected raw EMG signals to remove noise and obtain the linear envelope. The RMS, peak, and average EMG values were then calculated for each angle orientation and were compared to successfully determine which arm orientation produced the greatest muscle activation.

Results and Discussion

Peak Activation Analysis

As seen in Figure 1, the peak EMG amplitude recorded is for the 50° curl, followed by both the 0° and 45° angles.

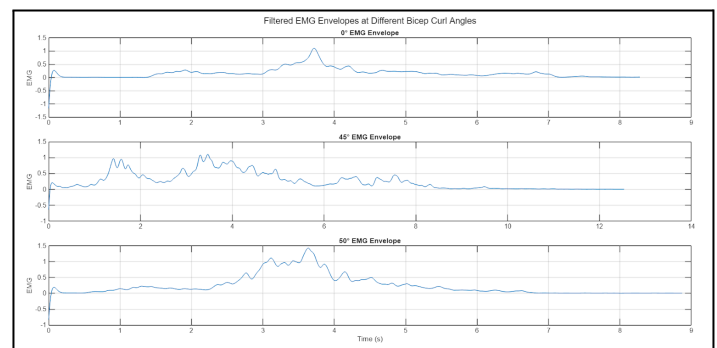


Figure 1. Filtered EMG envelopes of Different Bicep Curls

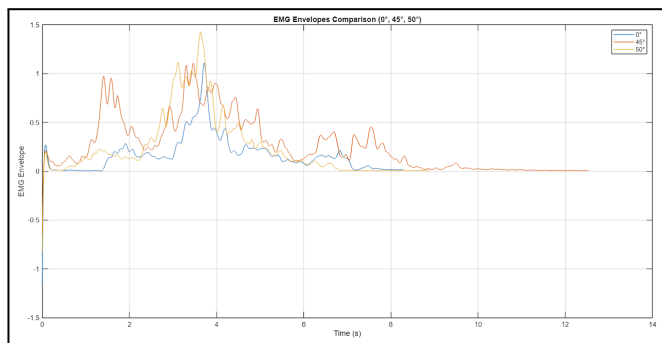


Figure 2. EMG envelopes comparison for all bicep exercises

This is quantitatively confirmed in Table 1, which shows the 50° curl yielded a Peak EMG of 1.4299, the highest recorded value. This spike in activation suggests that the muscle reached its highest instantaneous recruitment level at some point during the 50° movement.

Overall Effort (RMS and Mean EMG)

As seen in Table 1, the RMS value for the 50° bicep curl is the highest at 0.4001. EMG measures the electrical drive sent from the central nervous system (CNS) to the muscle. A high EMG RMS value translates to greater involvement of muscle fibers and increased activation of fast-twitch muscle fibers. RMS calculates the effective electrical power of the signal, providing a consistent measure of total neural drive that smooths out random noise spikes better than mean values. Muscle fibers are recruited from smallest to largest, and the largest fibers have the most growth potential, but they are only activated when demand is high. Therefore, a high EMG signal confirms that the muscle is successfully tapping into the specific fibers that have the capacity to grow significantly. While EMG signals are not directly related to hypertrophy, the recruitment of these high-threshold fibers makes EMG indirectly relevant to muscle growth.

It is important to note that the Mean EMG value was highest for the 45° angle (0.2632). This suggests that while the 50° curl caused the highest peak activation and overall force capacity (RMS), the 45° angle sustained a higher average level of moderate activation across the entire movement.

Conclusions

This study examined how shoulder angle influences the activation of the biceps brachii during a dynamic dumbbell

curl. The findings demonstrate that the 50 degree shoulder angle bicep curl produced the highest peak EMG (1.4299) and the highest RMS EMG (0.4001) values. This indicates that the 50 degree shoulder position maximizes the instantaneous muscle recruitment and overall effort of the biceps brachii during the exercise. This insight can be used to guide exercise selection for strengthening programs aimed at maximizing biceps activation.

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Table 1: Mean, Peak, and RMS EMG values for each bicep curl exercise.

Bicep Curl Angles/ degrees	Mean EMG	Peak EMG	RMS
0	0.1681	1.1089	0.2505
45	0.2632	1.1080	0.3774
50	0.2451	1.4299	0.4001

